Here is the complete project documentation for your event cost and registration calculation program, structured just like the previous ones!

1. Problem Statement, Objectives, and Assumptions

Problem Statement

Organizing institutional events, workshops, or symposiums requires meticulously tracking multiple expense categories (venue, equipment, catering, transport, certificates) against external financial backing (sponsorship). Calculating the precise number of student registrations required to cover any funding shortfall manually can lead to budgeting errors. An automated program is needed to evaluate financial gaps and determine the minimum student participation needed to break even.

Objectives

- **Automate Expense Aggregation:** To combine venue, equipment, catering, transport, and certificate expenses into a comprehensive total event cost.
- **Funding Gap Analysis:** To compute the net shortfall by subtracting sponsorship contributions from the total cost.
- **Break-Even Registration Calculation:** To dynamically calculate the minimum number of student registrations required to cover any remaining financial deficit, factoring in partial remainder division.

Assumptions

- Financial inputs (costs, sponsorship, fee per student) are valid numeric values.
- Student registration fees are structured as a flat rate per student.
- If total expenses are fully covered by sponsorship, the required student registration count is automatically set to zero.

2. Algorithm, Flowchart, and Pseudocode

Algorithm

1. **Start** the program.
2. **Input** venue, equipment, catering, transport, and certificates costs as floating-point numbers.
3. **Input** sponsorship amount and registration fee per student as floating-point numbers.
4. **Calculate Total Cost:** total_cost = venue + equipment + catering + transport + certificates.
5. **Calculate Shortfall:** shortfall = max(0, total_cost - sponsorship).

6. **Evaluate Minimum Registrations:**

   - If sponsorship >= total_cost, set minimum = 0.
   - Else:
     - Compute minimum = int(shortfall / fee).
     - If shortfall % fee != 0, increment minimum = minimum + 1 (ceiling division to account for partial student fees).

7. **Display Results:** Print Total Event Cost, Sponsorship Shortfall, and Minimum Registrations.

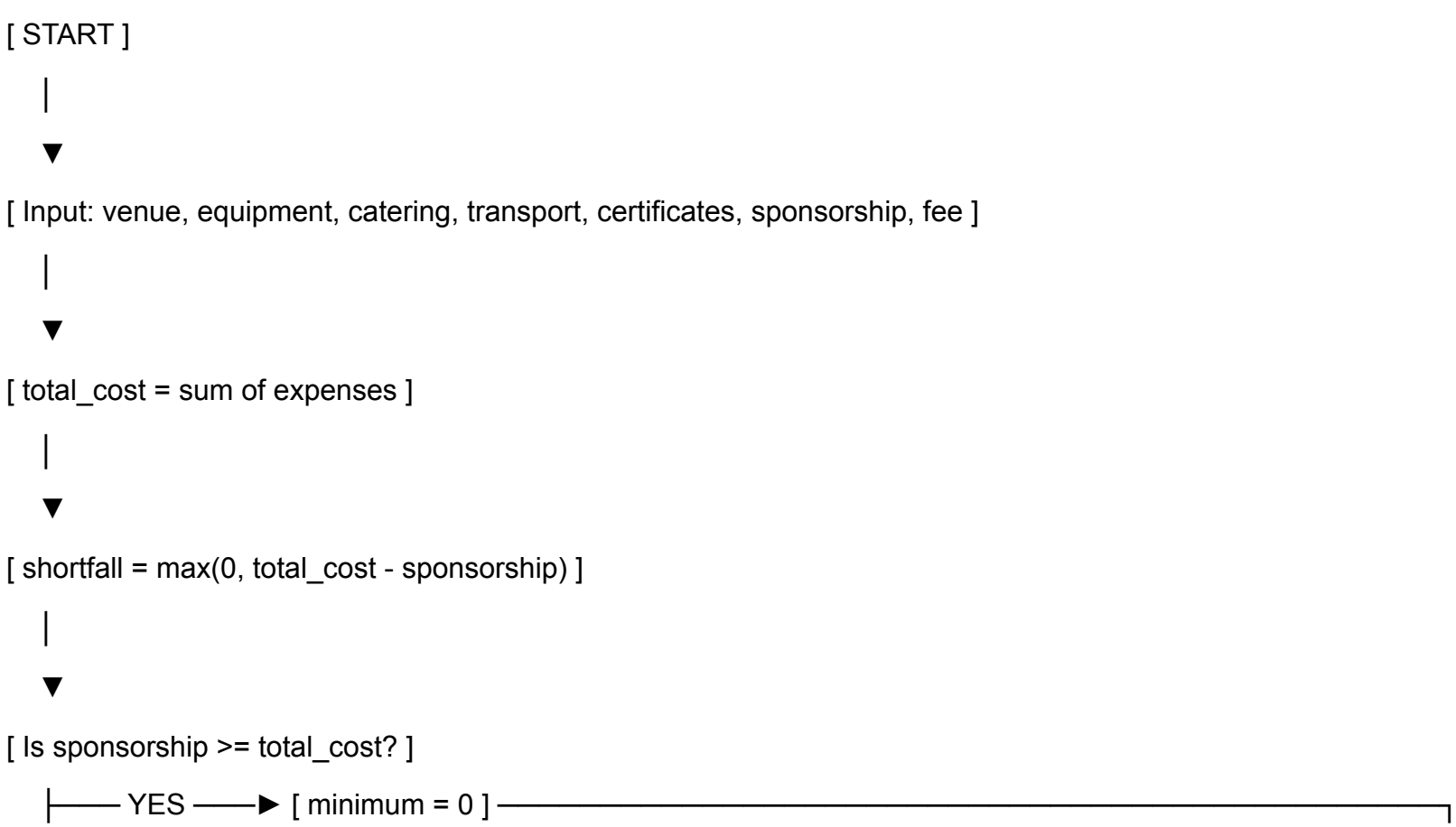
8. **Evaluate Coverage Status:**

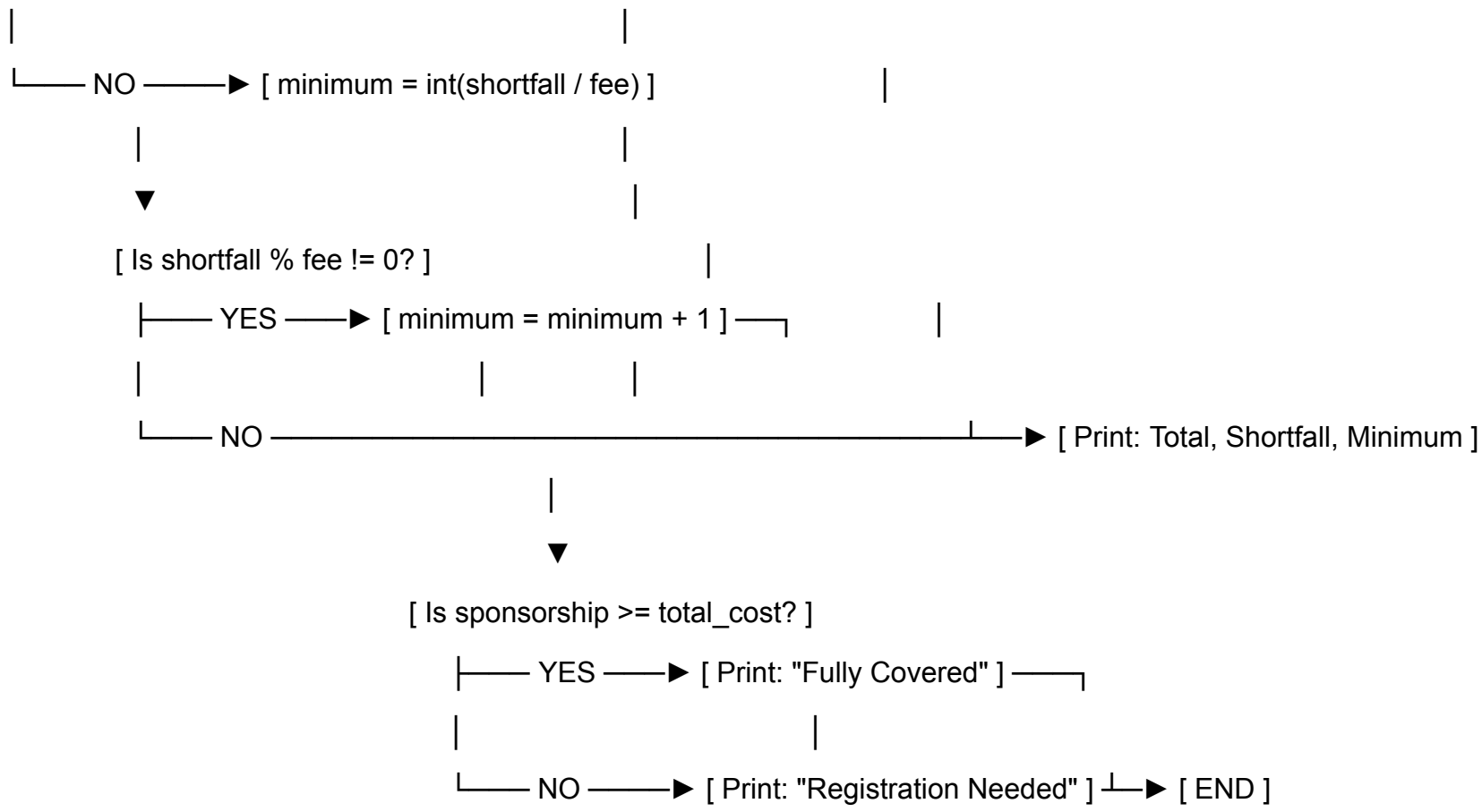
   - If sponsorship >= total_cost, print "Event Cost Fully Covered by Sponsorship".
   - Else, print "Registration Needed to Cover Remaining Cost".

9. **End** the program.

Flowchart Representation

Plaintext





Pseudocode

Plaintext

```
BEGIN
    PRINT "Enter venue cost: "
    READ venue
```

```
PRINT "Enter equipment cost: "
READ equipment
PRINT "Enter catering cost: "
READ catering
PRINT "Enter transport cost: "
READ transport
PRINT "Enter certificate cost: "
READ certificates

PRINT "Enter sponsorship amount: "
READ sponsorship
PRINT "Enter registration fee per student: "
READ fee

total_cost = venue + equipment + catering + transport + certificates
shortfall = max(0, total_cost - sponsorship)

IF sponsorship >= total_cost THEN
    minimum = 0
```

```
    ELSE
        minimum = INT(shortfall / fee)
        IF shortfall MOD fee != 0 THEN
            minimum = minimum + 1
        ENDIF
    ENDIF

    PRINT "Total Event Cost: Rs.", total_cost
    PRINT "Sponsorship Shortfall: Rs.", shortfall
    PRINT "Minimum Registrations:", minimum

    IF sponsorship >= total_cost THEN
        PRINT "Event Cost Fully Covered by Sponsorship"
    ELSE
        PRINT "Registration Needed to Cover Remaining Cost"
    ENDIF
END
```

3. Python Source Code, Test Cases, and Outputs

Python Source Code

Python

```
# Event Cost and Minimum Registration Calculator

venue = float(input("Enter venue cost: "))
equipment = float(input("Enter equipment cost: "))
catering = float(input("Enter catering cost: "))
transport = float(input("Enter transport cost: "))
certificates = float(input("Enter certificate cost: "))

sponsorship = float(input("Enter sponsorship amount: "))
fee = float(input("Enter registration fee per student: "))

total_cost = venue + equipment + catering + transport + certificates
shortfall = max(0, total_cost - sponsorship)

if sponsorship >= total_cost:
    minimum = 0
else:
    minimum = int(shortfall / fee)
```

```
    if shortfall % fee != 0:
        minimum = minimum + 1

print("Total Event Cost: Rs.", total_cost)
print("Sponsorship Shortfall: Rs.", shortfall)
print("Minimum Registrations:", minimum)

if sponsorship >= total_cost:
    print("Event Cost Fully Covered by Sponsorship")
else:
    print("Registration Needed to Cover Remaining Cost")
```

Test Cases Table

Test Case ID	Total Expenses (Rs.)	Sponsorship (Rs.)	Fee per Student (Rs.)	Shortfall (Rs.)	Minimum Registrations	Coverage Status	Status
TC_01	10000.0	12000.0	500.0	0.0	0	Event Cost Fully Covered by	Pass

						Sponsorship	
TC_02	15000.0	5000.0	250.0	10000.0	40	Registration Needed to Cover Remaining Cost	Pass
TC_03	8000.0	2000.0	300.0	6000.0	20	Registration Needed to Cover Remaining Cost	Pass
TC_04	12500.0	5000.0	400.0	7500.0	19	Registration Needed to Cover Remaining Cost	Pass

Simulated Terminal Output

Bash

Enter venue cost: 5000

Enter equipment cost: 3000

Enter catering cost: 4000

Enter transport cost: 1500

Enter certificate cost: 1500

Enter sponsorship amount: 5000

Enter registration fee per student: 250

Total Event Cost: Rs. 15000.0

Sponsorship Shortfall: Rs. 10000.0

Minimum Registrations: 40

Registration Needed to Cover Remaining Cost

4. Conclusion

This Python script effectively automates event financial planning by aggregating multiple expense categories, calculating external funding gaps, and executing ceil-division mathematical logic to determine precise participant break-even points. It provides an efficient and reliable programmatic tool for event organizers, academic coordinators, and club managers to ensure financial stability before hosting events.